



Utilising Fragrant Roots to Purify Wastewater From Fish Habitats and Reintroducing It to the Pond

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Abstract: Greywater waste is a type of wastewater that comes from bathroom water, washing that enters the drainage system, and is supplemented with rainwater whose downstream flow enters the river. This research aims to enhance the effectiveness of greywater treatment by using a physical model comprising a treatment process, namely a cattious wetland filter. This study applied the vetigenetic wetland method with and without media. The observed research variables were wastewater discharge variations, wetland residence time at intervals of 2 days, 4 days and 7 days, and the number of vetinous root stems, as well as water quality parameters (Y) consisting of pH, odor, color, turbidity, total dissolved solids (TDS), nitrate (NO_3^-), nitrite (NO_2^-), fecal coliform, and iron (Fe). The data analysis was based on descriptive evaluation of water quality parameters under different operational conditions, including variations in hydraulic retention time, flow rate, and plant density. The study is expected to show that using a physical model of a media-filtration root wetland can improve wastewater quality, with the output meeting clean-water quality standards that will be used as input to the pond. The qualified water quality consists of pH (7.9), NO_3^- 2.319 mg/L, NO_2^- 0.0922 mg/L, Fe 0.0286 mg/L, and the unqualified water quality consists of: odor 0.77, color 10.6 true colour units (TCU), turbidity 5.44 nephelometric turbidity units (NTU), TDS 175 mg/L, fecal coliform 85 colony-forming units (CFU)/100ml. The output comprises five parameters that do not meet treated effluent quality standards and will be used as inputs to the pond; therefore, further research on phytoremediation is warranted. A constructed wetland using vetiver grass effectively cleans fishpond greywater and reduces pollutant levels. This means it could be used as an initial or final step in water treatment when reuse is limited. However, in this study, the treated water didn't always meet all health standards. This suggests extra treatment steps are needed, along with more real-world testing to ensure safety.

Keywords: Constructed wetland; Vetiveria zizanioides; Greywater treatment; Fish pond wastewater; Phytoremediation; Water reuse

1 Introduction

Domestic wastewater is among the most significant sources of pollution in aquatic environments, particularly rivers [1]. Approximately 57% of river pollution originates from domestic wastewater discharges [2, 3]. River water, however, remains an essential resource for various purposes, including agricultural and landscape irrigation, washing vehicles and equipment, and other non-potable applications [4–7]. The discharge of untreated domestic wastewater deteriorates water quality and negatively affects plant growth and agricultural productivity when contaminated water is used for irrigation [8–11]. Given the increasing volume of domestic wastewater and its high pollution potential, the development of effective and sustainable wastewater treatment technologies has become a priority. Numerous studies have proposed simple and practical treatment technologies capable of improving wastewater quality through different treatment approaches [2–5, 11].

One of the most promising environmentally friendly approaches is the application of aquatic plants as phytoremediation agents [12, 13]. Constructed wetlands planted with aquatic vegetation remove pollutants through

a combination of physical filtration, sedimentation, microbial degradation, and plant uptake [14]. Among various aquatic plants, Vetiver grass (*Vetiveria zizanioides*), locally known as fragrant root grass, has attracted considerable attention because of its rapid growth, extensive root system, and wide availability [15]. Besides its traditional use for handicrafts and ornamental purposes, vetiver has demonstrated excellent capability in removing pollutants from wastewater [16–18]. These characteristics make vetiver a promising plant for producing treated effluent suitable for non-potable reuse, particularly in urban environments.

This study focuses on the treatment of domestic greywater to increase the availability of treated effluent for urban water reuse. Greywater refers to wastewater generated from sinks, kitchens, bathrooms, and laundry activities, excluding wastewater containing human excreta [19]. Greywater accounts for approximately 70–75% of the total domestic wastewater generated in households [20]. In many urban areas, this wastewater is discharged directly into drainage systems and eventually reaches nearby rivers without adequate treatment [21]. Consequently, a substantial amount of potentially reusable water is wasted, contributing to environmental degradation.

Integrated Water Resources Management (IWRM) promotes a coordinated approach that integrates environmental, social, and economic considerations in sustainable water-resource management [22]. Effective implementation of IWRM requires balancing stakeholder interests and multiple sustainability objectives [23].

In this study, water quality was evaluated using nine parameters: pH, odour, colour, turbidity, total dissolved solids (TDS), nitrate (NO_3^-), nitrite (NO_2^-), fecal coliform, and iron (Fe). The physicochemical and microbiological parameters were analyzed using standard water and wastewater examination procedures [24–33]. The treated water quality was evaluated according to the Indonesian Minister of Health Regulation No. 2 of 2023 [34]. This study proposes a simple and cost-effective constructed wetland model that can improve greywater quality using vetiver plants. Owing to its low cost, simple design, and ease of operation, the proposed system has strong potential for application in household-scale and community-based wastewater treatment.

Constructed wetlands have received considerable attention as environmentally friendly and sustainable wastewater treatment technologies [20]. These systems mimic natural wetlands by using vegetation, soil media, and microorganisms to remove contaminants from wastewater. Previous studies have demonstrated that vetiver plants effectively remove various pollutants, including dissolved nitrogen, phosphorus, sulfate, and heavy metals such as arsenic (As) and cadmium (Cd). Dyamanagowdru and Lokeshappa [20] reported that constructed wetlands planted with *Vetiveria zizanioides* and *Canna siamensis* achieved high pollutant removal efficiencies at hydraulic retention times of four and six days. Their study reported removal efficiencies of 96% for turbidity, 29% for TDS, 65% for total suspended solids (TSS), 70% for biochemical oxygen demand (BOD), 80% for chemical oxygen demand (COD), 53% for chloride, 50% for nitrate, and 55% for phosphate, demonstrating the effectiveness of vetiver-based constructed wetlands for wastewater treatment.

Previous research [3] reviewed different types of constructed wetlands, including free-water surface flow wetlands, subsurface-flow wetlands, hybrid systems, and floating-treatment wetlands, to identify suitable wastewater treatment technologies for developing countries. The study concluded that hybrid constructed wetlands provide superior treatment performance because they effectively reduce TSS and COD by 93% and 85.65%, respectively. In addition, *Typha angustifolia* was identified as an effective plant species for improving wastewater quality in constructed wetland systems.

Another study [35] evaluated a full-scale subsurface-flow constructed wetland installed in a residential area. The treatment system consisted of a wastewater filtration tank, a storage tank, and a constructed wetland measuring $2 \times 1 \times 0.65 \text{ m}^3$ with a treatment capacity of approximately $0.6 \text{ m}^3/\text{day}$. The wetland was planted with iris species using sand, gravel, and recycled plastic bottles as supporting media. The results showed that a hydraulic retention time of three days significantly improved wastewater quality, particularly by reducing BOD, compared with untreated influent wastewater.

Although numerous studies have demonstrated the effectiveness of constructed wetlands using vetiver and other aquatic plants, most have focused on domestic or industrial wastewater. Limited attention has been given to greywater generated from fishpond operations, despite its considerable environmental impact. Furthermore, few studies have systematically evaluated the combined effects of hydraulic retention time, flow rate, and plant density using a laboratory-scale physical model. Therefore, this study addresses these research gaps by investigating the performance of a vetiver-based constructed wetland for treating fishpond greywater under different operational conditions. The findings are expected to contribute to the development of simple, low-cost, and nature-based wastewater treatment technologies suitable for decentralized water reuse applications.

2 Literature Review

2.1 Wastewater

Waste is discarded material from human activities that has lost its value or utility. There are two types of liquid waste: industrial and household. Liquid waste from household activities typically consists solely of organic matter,

making it amenable to simple treatment to remove pollutants [29]. Liquid waste is water that contains organic pollutants or comes from:

- (1) Household activities, such as bathing, washing clothes, using the toilet, and cooking.
- (2) Commercial activities such as hotels, resorts, villas, vacation homes, offices, and supermarkets that produce liquid waste.
- (3) Industrial activities that produce wastewater, such as liquid waste from industries, agriculture, livestock farming, etc.
- (4) Rainwater that falls through rooftops, drainage channels, and road surfaces. Based on the nature of the waste, the wastewater treatment process can be divided into three categories [30]:

- (a) Physical process

This process is carried out mechanically without the addition of chemicals. It involves filtration, sedimentation, and flotation.

- (b) Chemical process

This process is carried out by utilising chemicals to neutralise pollutants.

- (c) Biological processes

This process removes pollutants using microorganisms. In practice, this treatment process does not operate independently but is often implemented alongside other methods.

2.2 Conceptual Framework of the Research

To clarify how the variables in this study interact, a conceptual framework was developed based on the study's problems and objectives. The variables in this study include the following [31]:

- 1) Free variables:

- (a) Number of fragrant root stems (50, 100, 150)
- (b) Number of discharges (1000, 1500, 2000) litres/day
- (c) Duration of stay (2 days, 4 days, 7 days), based on the type of plant used in the wetland, namely Vetiver (*Vetiveria zizanoides*)

- 2) The selected parameters, namely as follows:

- (a) pH
- (b) Odor
- (c) Color
- (d) TDS
- (e) Turbidity
- (f) NO_3^-
- (g) NO_2^-
- (h) Fecal coliform
- (i) Fe

3 Method

3.1 Research Location

This research was conducted at home, and the test results were then sent to the Jasa Tirta Malang Laboratory for further testing. This activity included collecting tilapia wastewater samples, analysing the initial and final water, and analysing the water output from the wetland. This physical model was made on a scale of 1:150, adjusting to the available space in the house, including:

- (a) Water Reservoir
- (b) Filter 2 wetland filter boxes

Figure 1 explains the preparations and tools before observation, such as:

- (a) Sampling from the research location into bottles
- (b) Tilapia pond
- (c) Fragrant root plants with residence time, variation in the number of fragrant root stems, and variation in flow rate
- (d) Output with variations in days, discharge, and number of fragrant root stems

3.2 Waste Treatment Model

To achieve treated effluent quality standards, research was conducted using the model illustrated in Figure 2, which utilises soil as a medium for wetland plants. All treatments are similar to phytoremediation, except that these use soil submerged in water.

Description of the Waste Treatment Process in Figure 2, such as:

- 1) Treated effluent as input entering the tilapia pond

- 2) Tilapia pond
- 3) Tilapia pond wastewater
- 4) Fragrant root wetlands (model scale) that use tilapia pond wastewater
- 5) The output from wetlands that use wastewater is treated effluent that will re-enter the tilapia pond.

Wetlands are areas where the soil is saturated with water, either permanently or seasonally. These areas are sometimes partially or completely flooded by shallow water and are classified as wetlands, including swamps (e.g., mangrove swamps), marshes, and peatlands. The water that floods wetlands can be classified as freshwater, brackish water, or saltwater. In this case, we work using a scale model in the laboratory (see Figure 2).



Figure 1. Preparations and tools: (a) sampling; (b) model scale; (c) fragrant root plants; and (d) output results

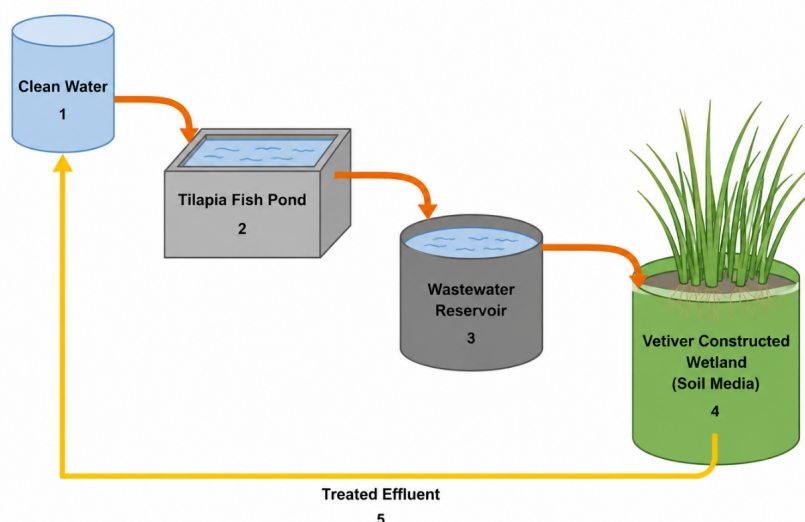


Figure 2. Treated effluent treatment process, use of ponds, discharge, and processing to produce treated effluent again

All the greywater used in this study was sourced from the same fish pond and freshly collected for each experiment. Before each new run, the water was mixed to maintain consistency for that particular test. Water quality was not monitored over time; the study focused only on the water used in each setup. At the beginning of each experiment, the primary water quality parameters were measured to document the initial conditions. Since each test was conducted

only once without replication, the influent values were presented as single data points for each scenario rather than as averages. These values allow me to compare how the treatment performed under various retention times, flow rates, and plant densities.

Standard analytical methods were used to assess water quality, following typical lab procedures. Turbidity, pH, and TDS were all measured using calibrated portable meters and for Fecal coliforms, established culture-based methods were used. However, this study does not assess measurement uncertainty or instrument precision, which likely contributed to the variability of the results. In the future, it would be sensible to include detection limits for each method and perform replicate measurements to enhance the analysis.

3.3 The Research Model Location

Given the study's general focus on domestic wastewater treatment in urban areas, it is necessary to provide an overview of the locations where the technology will be applied. This community service project is located in a housing complex in Malang City, East Java, namely Pondok Harapan Indah (Poharin). Specifically, this housing complex is located on Jalan Terusan Sigura-gura, Karangbesuki Village, Sukun District, Malang City. This location was chosen because the housing complex is quite old, having been established in 1981. This indicates that population density is relatively high, resulting in substantial domestic wastewater generation. So far, there has been no adequate greywater treatment facility in Poharin. Still, at least residents have initiated the construction of fish ponds to utilise greywater (see Figure 3). This idea has motivated researchers to support domestic wastewater treatment initiatives.



Figure 3. Research location

3.4 Experimental Replication and Data Reporting

Each experimental condition was conducted only once, due to time and resource constraints, in a controlled laboratory environment. The experiments were not repeated, so the figures in the tables are from single trials rather than averages over several runs. These data illustrate how retention time, flow rate, and plant density affect outcomes, but they are not suitable for drawing broad statistical conclusions. Thus, to present the observed patterns rather than make generalised statements.

4 Results and Discussions

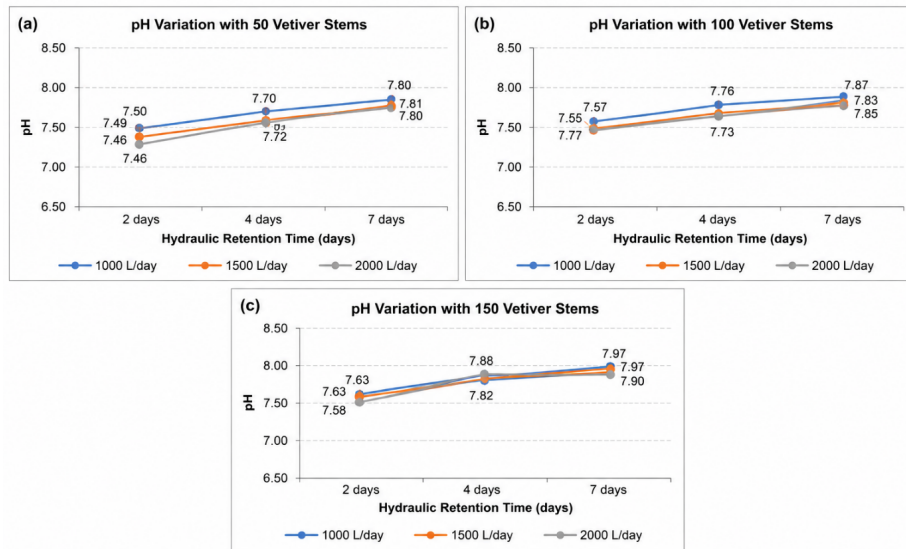
This section presents the results for water quality parameters, including pH, color, turbidity, TDS, NO_3^- , NO_2^- , Fe, odor, and fecal coliforms. Water quality is influenced by water discharge, the number of fragrant roots, and retention time. The test results are presented using line graphs, as shown below. The results presented in this section correspond to single experimental runs for each treatment condition and are intended to demonstrate comparative performance trends among the tested scenarios. The influent water quality values reported for each test condition represent initial measurements taken at the start of each experimental run and are intended to characterize the input conditions rather than capture temporal variability.

4.1 The pH Parameter

Table 1 shows that, with 50 fragrant root stems and each increased with the addition of water, from 7.30 to 7.80 across different retention periods. However, Figure 4 shows that, with 100 fragrant root stems and each additional water addition, the pH value increases from 7.30 to 7.85 across different retention periods. Meanwhile, the results for the pH parameter indicate that, with 150 fragrant root stems and each addition of water, the pH value increases from 7.30 to 7.90 across different retention periods.

Table 1. Results of greywater waste treatment with submerged soil media pH (6.5–8.5)

Number of Vetiver Stems	Water Discharge (L/day)	Greywater (Initial pH)	2 Days	4 Days	7 Days
50	1000	7.30	7.50	7.70	7.80
50	1500	7.30	7.49	7.68	7.81
50	2000	7.30	7.46	7.72	7.80
100	1000	7.30	7.55	7.76	7.87
100	1500	7.30	7.55	7.73	7.83
100	2000	7.30	7.57	7.73	7.85
150	1000	7.30	7.60	7.84	7.93
150	1500	7.30	7.63	7.82	7.97
150	2000	7.30	7.58	7.88	7.90

**Figure 4.** Effect of hydraulic retention time on pH at different vetiver plant densities: (a) 50 vetiver stems; (b) 100 vetiver stems; and (c) 150 vetiver stems.

4.2 The Odor Parameter

The description of the odor parameters shows that with 50 bars of fragrant roots and each addition of water flow at different retention periods, the odor value can be reduced from 2 to 0.97, as shown in Table 2 and Figure 5. The results of the odor parameter description show that, with 100 fragrant root stems and each addition of water discharge at different retention periods, the odor value can be reduced from 2 to 0.91. The results of the odor parameter description indicate that, with 150 fragrant root stems and each addition of water discharged at different retention periods, the odor value can be reduced to 0.77.

Table 2. Results of greywater treatment using a constructed wetland with soil media: odor parameter (water quality standard: odorless)

Number of Vetiver Stems	Water Discharge (L/Day)	Initial Odor Score	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	2.00	2.00	2.00	1.00
	1500	2.00	2.00	2.02	1.04
	2000	2.00	1.92	2.06	0.97
100	1000	2.00	1.86	1.90	0.90
	1500	2.00	1.79	1.98	0.95
	2000	2.00	1.90	1.96	0.91
150	1000	2.00	1.75	1.73	0.81
	1500	2.00	1.70	1.76	0.80
	2000	2.00	1.68	1.69	0.77

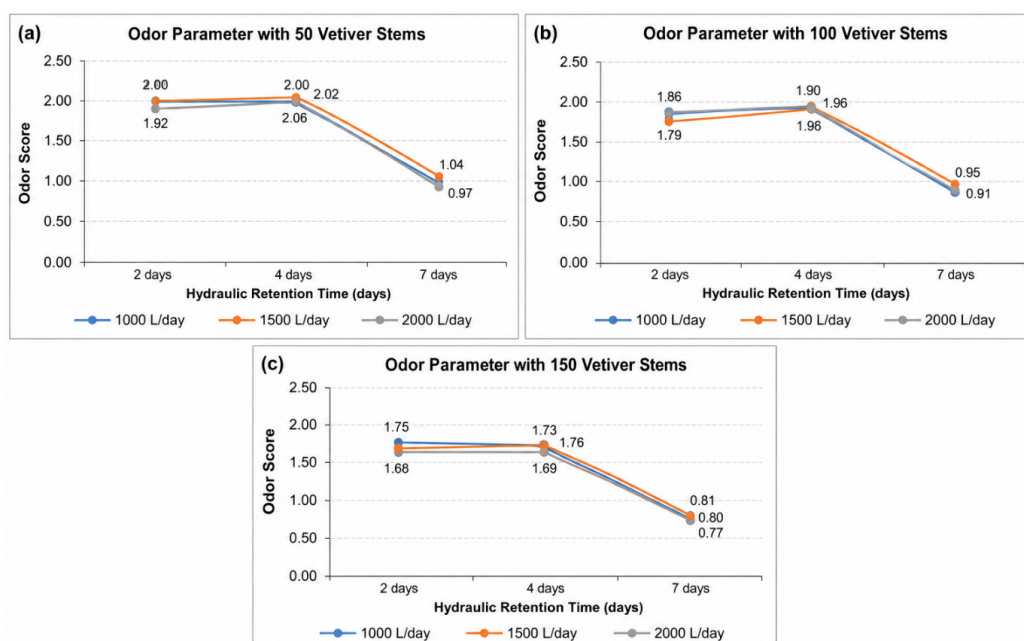


Figure 5. Effect of hydraulic retention time on the odor parameter at different vetiver stem densities: (a) odor parameter with 50 vetiver stems; (b) odor parameter with 100 vetiver stems; and (c) odor parameter with 150 vetiver stems

4.3 The Color Parameter

The results of the color parameter description show that with 50 fragrant root stems and each addition of water discharge at different retention periods, the color value can be reduced from 34.63 to 11.9 true colour units (TCU), as shown in Table 3 and Figure 6. The results of the color parameter description show that, with 100 fragrant root stems and each additional water discharge at different retention periods, the color value can be reduced from 34.63 to 11.5 TCU. The results of the color parameter description show that with 150 fragrant root stems and each addition of water discharge at different retention periods, the color value can be reduced from 34.63 to 10.6 TCU.

Table 3. Results of greywater treatment using a constructed wetland with soil media: color parameter (water quality standard: ≤ 10 TCU)

Number of Vetiver Stems	Water Discharge (L/Day)	Initial Color (TCU)	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	34.63	30.0	20.0	12.0
	1500	34.63	30.0	19.8	12.0
	2000	34.63	28.5	20.2	11.9
100	1000	34.63	27.6	18.2	11.3
	1500	34.63	27.6	18.4	10.7
	2000	34.63	29.0	17.7	11.5
150	1000	34.63	25.4	16.4	10.4
	1500	34.63	24.4	15.6	10.8
	2000	34.63	26.7	16.2	10.6

Note: TCU = true colour units.

4.4 The Total Dissolved Solids Parameter

The TDS parameter description results show that, with 50 bars of fragrant roots and each addition of water, the TDS value decreases from 332 to 192 mg/L, as shown in Table 4 and Figure 7. The results of the TDS parameter description show that, with 100 fragrant root stems and each additional water discharge at different retention periods, the TDS value can be reduced from 332 to 179 mg/L. The results of the TDS parameter description indicate that, with 150 fragrant root stems and each water discharge at different retention periods, the TDS value can be reduced from 332 to 175 mg/L.

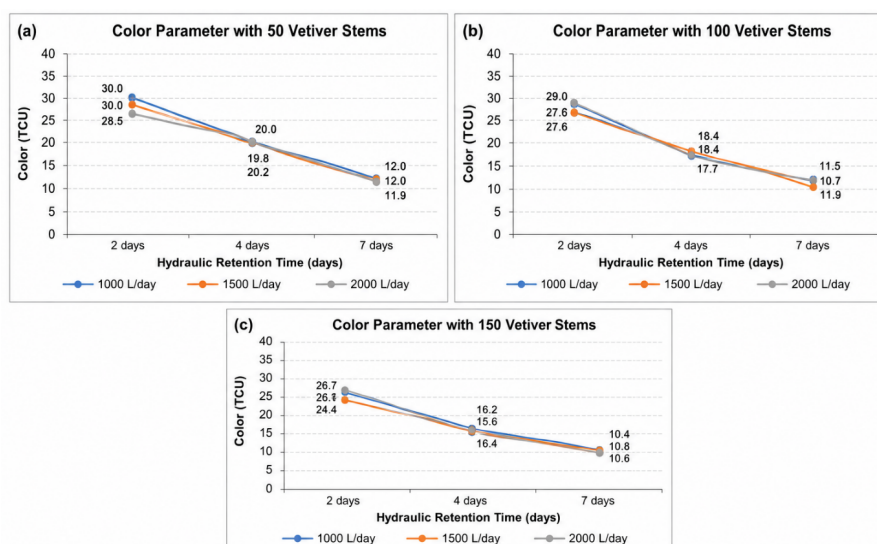


Figure 6. Effect of hydraulic retention time on the color parameter in greywater treated with different vetiver stem densities: (a) color parameter with 50 vetiver stems; (b) color parameter with 100 vetiver stems; (c) color parameter with 150 vetiver stems

Table 4. Results of greywater treatment using a constructed wetland with soil media parameter

Number of Vetiver Stems	Water Discharge (L/Day)	Initial TDS (mg/L)	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	332	300	250	200
	1500	332	306	263	200
	2000	332	303	243	192
100	1000	332	279	233	188
	1500	332	290	233	196
	2000	332	290	223	179
150	1000	332	254	212	175
	1500	332	267	207	178
	2000	332	241	207	175

Note: TDS = total dissolved solids; water quality standard: < 300 mg/L.

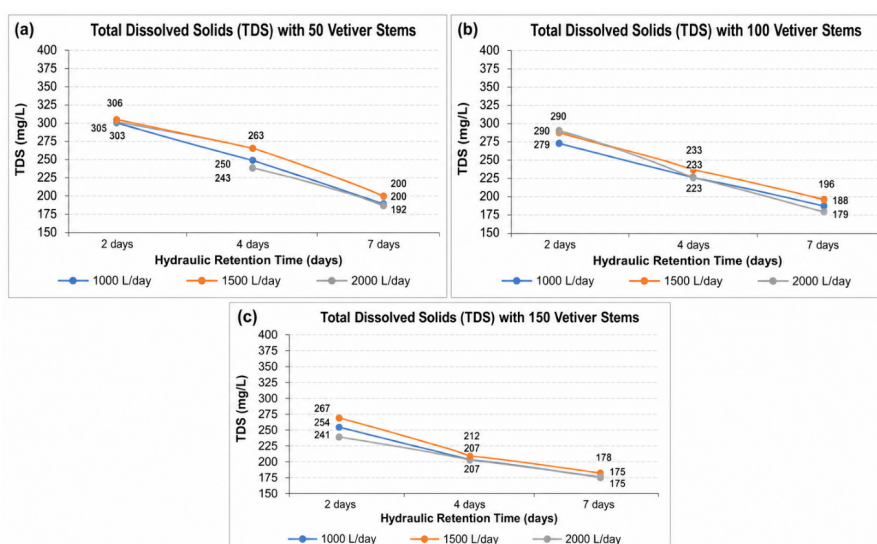


Figure 7. Effect of hydraulic retention time on total dissolved solids (TDS) in greywater treated with different vetiver stem densities: (a) TDS with 50 vetiver stems; (b) TDS with 100 vetiver stems; and (c) TDS with 150 vetiver stems

4.5 The Turbidity Parameter

The results of the turbidity parameter description indicate that, with 50 fragrant root bars and each water discharge addition at different retention periods, the turbidity value can be reduced from 119 to 6.42 nephelometric turbidity units (NTU), as shown in Table 5 and Figure 8. The results of the turbidity parameter description show that, with 100 fragrant root stems and each additional water discharge at different retention periods, the turbidity value can be reduced from 119 to 5.68 NTU. The results of the turbidity parameter description indicate that, with 150 fragrant root stems and each additional water discharge at different retention periods, the turbidity value can be reduced from 119 to 5.44 NTU.

Table 5. Results of greywater treatment using a constructed wetland with soil media: turbidity parameter

Number of Vetiver Stems	Water Discharge (L/Day)	Initial Turbidity (NTU)	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	119.00	67.00	20.00	6.36
	1500	119.00	69.68	20.20	6.30
	2000	119.00	63.65	20.40	6.42
100	1000	119.00	60.30	19.00	5.85
	1500	119.00	61.51	18.81	5.91
	2000	119.00	60.30	18.43	5.68
150	1000	119.00	54.27	17.48	5.38
	1500	119.00	53.73	17.65	5.11
	2000	119.00	54.27	18.00	5.44

Note: NTU = nephelometric turbidity units; water quality standard: <3 NTU.

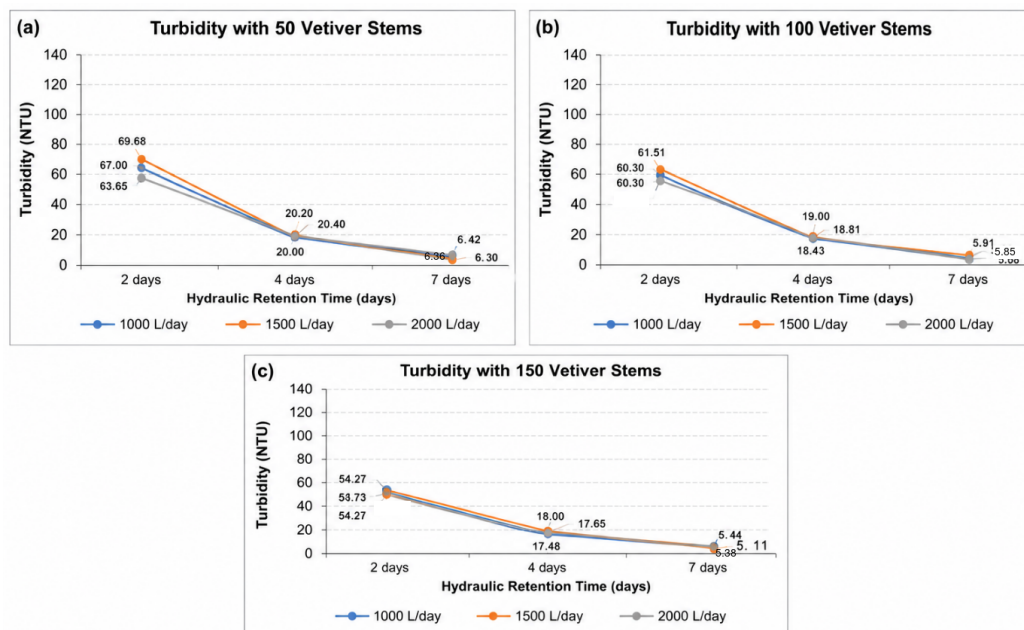


Figure 8. Effect of hydraulic retention time on the turbidity parameter in greywater treated with different vetiver stem densities: (a) turbidity with 50 vetiver stems; (b) turbidity with 100 vetiver stems; and (c) turbidity with 150 vetiver stems

4.6 The Nitrate Parameter

The description of the NO_3^- parameter indicates that, with 50 fragrant root stems and each addition of water, and with 100 fragrant root stems and each additional water discharge at different retention periods, as shown in Table 6 and Figure 9. For 150 fragrant root stems and each additional water discharge at different retention periods, the NO_3^- concentration increased from 1.412 mg/L to 2.319 mg/L.

Table 6. Results of greywater treatment using a constructed wetland with soil media: nitrate (NO_3^-) parameter (water quality standard: ≤ 20 mg/L)

Number of Vetiver Stems	Water Discharge (L/Day)	Initial Nitrate (mg/L)	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	1.412	6.950	5.764	2.734
	1500	1.412	6.742	5.937	2.679
	2000	1.412	6.603	5.533	2.652
100	1000	1.412	6.603	5.303	2.597
	1500	1.412	6.470	5.462	2.545
	2000	1.412	6.603	5.356	2.467
150	1000	1.412	6.272	4.773	2.415
	1500	1.412	5.959	4.963	2.319
	2000	1.412	6.398	4.820	2.319

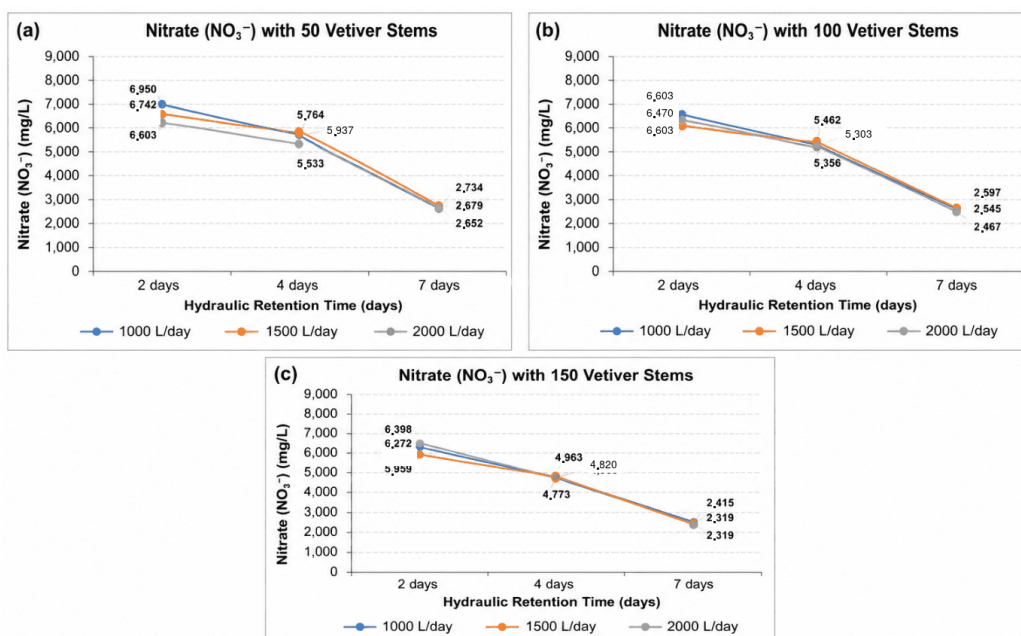


Figure 9. Effect of hydraulic retention time on the nitrate (NO_3^-) parameter in greywater treated with different vetiver stem densities: (a) nitrate (NO_3^-) with 50 vetiver stems; (b) nitrate (NO_3^-) with 100 vetiver stems; and (c) nitrate (NO_3^-) with 150 vetiver stems

4.7 The Nitrite Parameter

Results for the NO_2^- parameter indicate that, as shown in Table 7 and Figure 10, nitrite performance varied for 50 fragrant root stems, 100 fragrant root stems, and 150 fragrant root stems under different water-discharge and retention-period conditions. For 150 fragrant root stems, the NO_2^- concentration increased from 0.0086 mg/L to 0.0922 mg/L.

4.8 The Fecal Coliform Parameter

The results of the fecal coliform parameter description show that with 50 fragrant root stems and each addition of water discharge at different retention periods, the fecal coliform value can be reduced from 260 to 101 colony-forming unit (CFU)/100ml, as shown in Table 8 and Figure 11. The results of the fecal coliform parameter description show that with 100 fragrant root stems and each addition of water discharge at different retention periods, the fecal coliform value can be reduced from 260 to 91 CFU/100ml. The description of Fecal coliform parameters shows that with 150 fragrant root stems and each addition of water, the Fecal coliform value can be reduced from 260 to 85 CFU/100ml.

4.9 Iron Parameter

The results for the Fe parameter indicate that, for 50 fragrant root stems across different retention-period water discharges, the Fe concentration decreased from 0.0585 mg/L to 0.0265 mg/L, as shown in Table 9 and Figure 12.

For 100 fragrant root stems, the Fe concentration decreased from 0.0585 mg/L to 0.0265 mg/L under varying retention-period water discharges. For 150 fragrant root stems, the Fe concentration decreased from 0.0585 mg/L to 0.0286 mg/L.

Table 7. Results of greywater treatment using a constructed wetland with soil media: nitrite (NO_2^-) parameter (water quality standard: ≤ 3 mg/L)

Number of Vetiver Stems	Water Discharge (L/Day)	Initial Nitrite (mg/L)	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	0.0086	0.5319	0.3451	0.1023
	1500	0.0086	0.5159	0.3416	0.1064
	2000	0.0086	0.5159	0.3313	0.1023
100	1000	0.0086	0.4787	0.3175	0.0972
	1500	0.0086	0.4883	0.3238	0.0933
	2000	0.0086	0.4787	0.3302	0.1020
150	1000	0.0086	0.4356	0.2857	0.0904
	1500	0.0086	0.4269	0.2972	0.0940
	2000	0.0086	0.4226	0.2972	0.0922

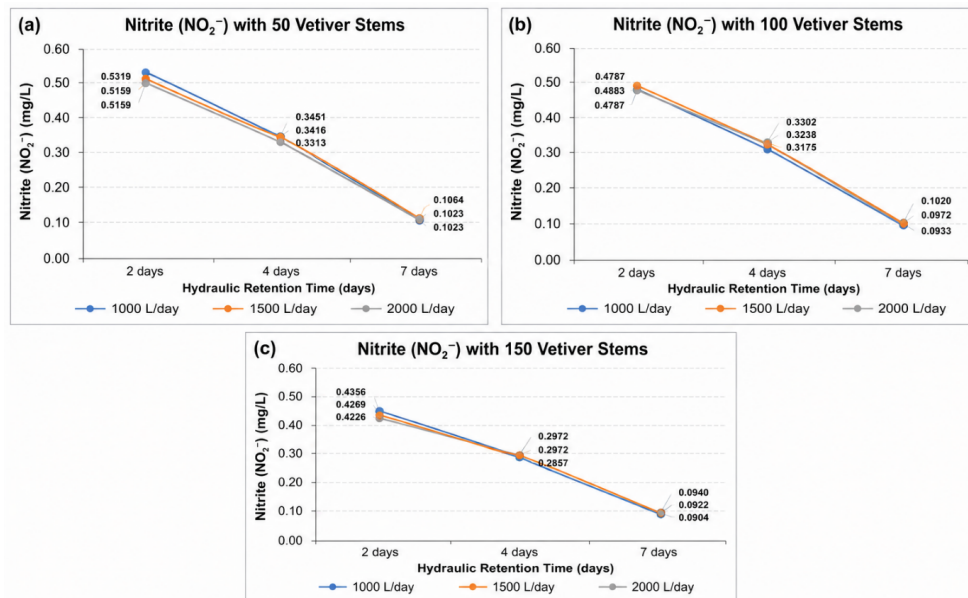


Figure 10. Effect of hydraulic retention time on the nitrite (NO_2^-) parameter in greywater treated with different vetiver stem densities: (a) nitrite (NO_2^-) with 50 vetiver stems; (b) nitrite (NO_2^-) with 100 vetiver stems; and (c) nitrite (NO_2^-) with 150 vetiver stems

Table 8. Results of greywater treatment using a constructed wetland with soil media: fecal coliform parameter

Number of Vetiver Stems	Water Discharge (L/Day)	Initial Fecal Coliform (CFU/100 mL)	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	260	250	180	100
	1500	260	250	180	100
	2000	260	250	180	101
100	1000	260	233	167	91
	1500	260	233	168	91
	2000	260	232	167	91
150	1000	260	214	151	86
	1500	260	213	151	86
	2000	260	214	150	85

Note: CFU = colony-forming unit; water quality standard: 0 CFU/100 mL.

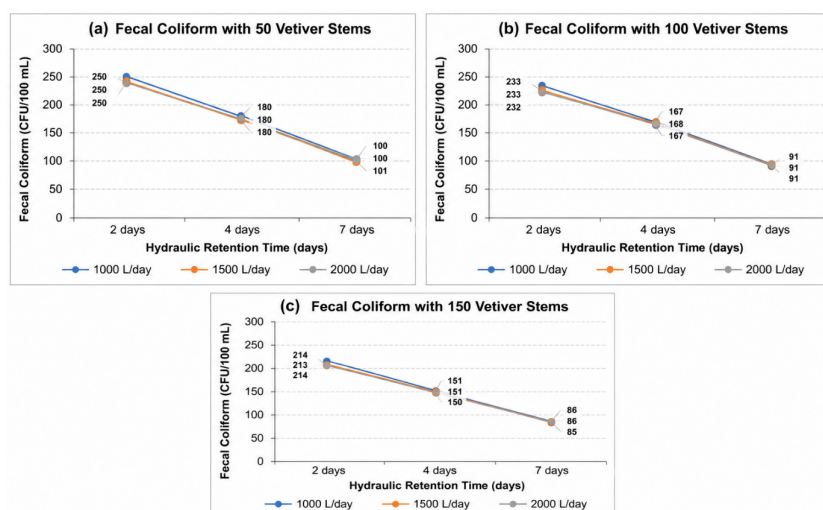


Figure 11. Effect of hydraulic retention time on the fecal coliform parameter in greywater treated with different vetiver stem densities: (a) fecal coliform with 50 vetiver stems; (b) fecal coliform with 100 vetiver stems; and (c) fecal coliform with 150 vetiver stems

Table 9. Results of greywater treatment using a constructed wetland with soil media: iron (Fe) parameter (water quality standard: ≤ 0.2 mg/L)

Number of Vetiver Stems	Water Discharge (L/Day)	Initial Iron (Fe) (mg/L)	Hydraulic Retention Time (Days)		
			2 Days	4 Days	7 Days
50	1000	0.0585	0.0432	0.0325	0.0260
	1500	0.0585	0.0454	0.0312	0.0255
	2000	0.0585	0.0432	0.0309	0.0265
100	1000	0.0585	0.0471	0.0351	0.0276
	1500	0.0585	0.0461	0.0344	0.0273
	2000	0.0585	0.0494	0.0362	0.0265
150	1000	0.0585	0.0499	0.0386	0.0289
	1500	0.0585	0.0494	0.0394	0.0286
	2000	0.0585	0.0519	0.0375	0.0286

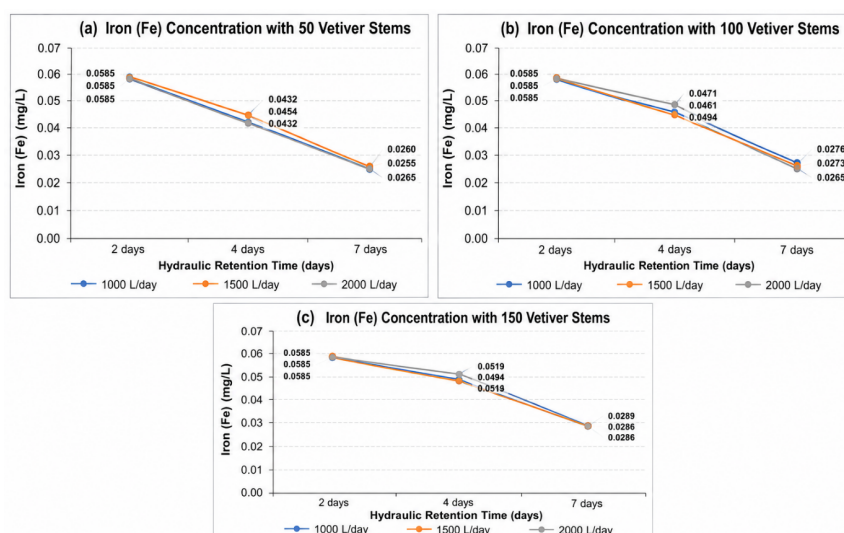


Figure 12. Effect of hydraulic retention time on the iron (Fe) parameter in greywater treated with different vetiver stem densities: (a) iron (Fe) concentration with 50 vetiver stems; (b) iron (Fe) concentration with 100 vetiver stems; and (c) iron (Fe) concentration with 150 vetiver stems

4.10 Discussion

In this study, vetiver-based constructed wetlands were evaluated as a nature-based solution for treating greywater from fish pond operations. The treatment system effectively reduced organic matter, turbidity, TDS, and fecal coliform concentrations, demonstrating the potential of vetiver-planted wetlands to improve greywater quality. Table 10 summarizes the treatment performance. These findings are consistent with previous studies on hybrid constructed wetlands and vetiver-based treatment systems, which reported effective removal of organic matter, suspended solids, and nutrients under various hydraulic loading conditions [7, 13]. With appropriate design and operation, constructed wetlands planted with macrophytes such as vetiver can significantly improve water quality, making treated greywater suitable for irrigation and other non-potable applications. Furthermore, constructed wetlands have been widely recognized as reliable, low-energy, and cost-effective technologies for decentralized wastewater treatment [8].

Table 10. Summary of treated greywater quality compared with the applicable treated effluent standards

No.	Parameter	Raw Greywater	Final Treated Effluent (50 Vetiver Stems, 7 Days)	Final Treated Effluent (100 Vetiver Stems, 7 Days)	Final Treated Effluent (150 Vetiver Stems, 7 Days)	Treated Effluent Standard	Unit
1	pH	5.30	7.80	7.85	7.90	6.5–8.5	-
2	Odor	2	0.97	0.91	0.77	Odorless	-
3	Color	34.63	11.9	11.5	10.6	≤10	TCU
4	TDS	332	192	179	175	<300	mg/L
5	Turbidity	119	6.42	5.68	5.44	<3	NTU
6	Nitrate (NO ₃ ⁻)	1.412	2.652	2.467	2.319	≤20	mg/L
7	Nitrite (NO ₂ ⁻)	0.0086	0.1023	0.1020	0.0922	≤3	mg/L
8	Fecal coliform	260	101	91	85	0	CFU/100 mL
9	Iron (Fe)	0.0585	0.0265	0.0265	0.0286	≤0.2	mg/L

Note: TDS = total dissolved solids; TCU = true colour units; CFU = colony-forming unit; NTU = nephelometric turbidity units; “-” = no applicable unit.

The constructed wetland evaluated in this study was a laboratory-scale physical model, approximately 1:150 of a potential full-scale system. Laboratory-scale experiments are valuable for investigating treatment mechanisms and identifying critical operational parameters; however, they cannot fully replicate the hydraulic complexity of field-scale wetlands. Scaling down may alter flow distribution, hydraulic retention time, media packing characteristics, and root-zone interactions, resulting in hydraulic behavior that differs from that of full-scale systems. Consequently, the findings should be regarded as preliminary design guidance rather than direct predictions of field performance. Future pilot-scale and full-scale investigations are recommended to validate hydraulic performance, treatment efficiency, and long-term operational stability under real environmental conditions.

Beyond the measured improvements in water quality, the observed reductions in organic matter, nutrients, suspended solids, and microbial contaminants have important environmental implications. Implementation of this treatment system in residential communities or small settlements could substantially reduce pollutant discharge into rivers and drainage systems. Lower organic and nutrient loading would reduce eutrophication risks, improve dissolved oxygen concentrations, and enhance ecological conditions for aquatic organisms. Therefore, the significance of this study extends beyond pollutant removal efficiencies by demonstrating the potential contribution of nature-based wastewater treatment systems to sustainable environmental management.

The observed treatment performance can be explained through several complementary mechanisms, including physical filtration, sedimentation, microbial biodegradation within biofilms, and rhizosphere-mediated pollutant transformation. In subsurface-flow constructed wetlands, the filter media retain suspended particles while simultaneously providing attachment surfaces for biofilm development. Meanwhile, vetiver roots improve oxygen transfer within the rhizosphere, creating favorable microenvironments for microbial activity and nutrient transformation. Similar mechanisms have been reported in previous investigations demonstrating the effectiveness of vetiver-planted wetlands for degrading organic pollutants under controlled hydraulic loading conditions [9].

The present findings are also consistent with recent investigations of constructed wetlands for decentralized water reuse. Previous field and pilot-scale studies have demonstrated that constructed wetlands effectively improve greywater quality while substantially reducing conventional pollutants and fecal indicator bacteria under appropriate operating conditions [8, 9, 11]. Moreover, hybrid constructed wetlands have shown superior treatment performance by combining aerobic and anoxic treatment zones, thereby enhancing nutrient removal efficiency and improving

treatment stability under variable hydraulic conditions [3, 7].

An increase in nitrate or nitrite concentrations observed during certain treatment stages is likely associated with enhanced nitrification. Increased oxygen availability within the wetland media promotes the microbial oxidation of reduced nitrogen compounds into nitrate. This process commonly occurs as organic matter concentrations decline and aerobic conditions become more favorable [31]. Consequently, elevated nitrate concentrations should not necessarily be interpreted as treatment failure but rather as possible evidence of nitrogen transformation within the wetland. For a more comprehensive assessment of nitrogen removal, future studies should evaluate total nitrogen (TN) in addition to individual nitrogen species.

The reduction in fecal coliform concentrations further demonstrates the capability of vegetated constructed wetlands to reduce microbial contamination through multiple mechanisms, including physical filtration, microbial attachment to biofilms, predation by indigenous microorganisms, natural bacterial die-off, and unfavorable environmental conditions within the rhizosphere [31]. Nevertheless, although fecal coliform concentrations decreased considerably, the treated effluent should not be considered suitable for direct human consumption. Additional disinfection processes may still be required depending on local water reuse regulations and the intended end use. Recent reviews emphasize that pathogen removal in constructed wetlands may vary according to operational conditions and therefore recommend integrating wetlands within a multi-barrier treatment strategy to ensure microbiological safety [34, 36].

Influent water quality may also influence the interpretation of treatment efficiency. In short-term laboratory studies, variations in influent characteristics can substantially affect removal efficiencies because treatment performance is strongly dependent on initial pollutant concentrations. Therefore, future investigations should clearly report the influent sampling strategy, confirm whether influent samples originate from a homogeneous and well-mixed source, and present the mean values together with standard deviations for each experimental condition. Such information would improve the reproducibility of the study and facilitate comparisons with other decentralized greywater treatment systems.

A comparison between the present study and recent investigations of constructed wetlands for greywater treatment is presented in Table 11.

This study evaluated the performance of a laboratory-scale constructed wetland during relatively short hydraulic retention times under controlled experimental conditions. Therefore, the findings should be interpreted as evidence of short-term treatment effectiveness rather than confirmation of long-term operational reliability. The long-term performance of constructed wetlands may be influenced by several factors, including media clogging, plant growth and harvesting cycles, seasonal climatic variations, and changes in microbial communities. Recent long-term studies have demonstrated that treatment efficiency can vary considerably across seasons and different stages of plant development. Consequently, monitoring under both wet and dry seasonal conditions is necessary before the system can be considered suitable for continuous year-round operation [11].

Based on the present findings and previous studies, three practical recommendations can be proposed. First, influent flow and pollutant loading should be stabilized through simple equalization or sedimentation units before entering the constructed wetland. Such pretreatment can minimize hydraulic shock loading and improve treatment stability. Second, hydraulic retention time and vetiver planting density should be optimized to achieve an appropriate balance between physical filtration and biological pollutant removal, as supported by previous pilot-scale studies and reviews on constructed wetlands for water reuse [9, 13, 20]. Third, where higher microbiological safety is required, constructed wetlands should be integrated with additional treatment processes, such as disinfection, as part of a multi-barrier treatment strategy for reclaimed water reuse [34, 36].

The results indicate that vetiver-based constructed wetlands represent a promising low-energy and environmentally sustainable technology for polishing greywater generated from fish habitat facilities. However, because the present investigation was conducted over a relatively short experimental period, additional pilot-scale and long-term studies are required to evaluate treatment stability, hydraulic performance, seasonal variability, media longevity, and long-term operational efficiency.

From both policy and practical perspectives, the proposed treatment system supports the increasing adoption of decentralized wastewater management, particularly in rural areas, peri-urban communities, and locations where centralized wastewater treatment infrastructure is unavailable or economically impractical. The findings provide baseline design information regarding hydraulic retention time, vetiver planting density, and expected treatment performance that may assist engineers, planners, and decision-makers in implementing nature-based wastewater treatment technologies. Furthermore, this study supports the integration of constructed wetlands into sustainable water resource management policies aimed at reducing diffuse water pollution. Accordingly, the treated effluent should be regarded as a valuable water resource suitable for restricted non-potable reuse, such as landscape irrigation and environmental applications, rather than as water intended for unrestricted human consumption.

Table 11. Comparison of the present study with recent studies on constructed wetlands for greywater treatment

Study	Wastewater Type	Wetland Configuration	Plant Species	Hydraulic Retention Time (HRT)	Main Findings	Key Remarks
Present study	Greywater from fish habitat (tilapia pond)	Subsurface constructed wetland with soil media (laboratory scale)	Vetiveria zizanioides	2, 4, and 7 days	Significant reductions in TDS, Fe, fecal coliform, turbidity, and color	Suitable as a low-cost polishing treatment for non-potable water reuse
Dell’Osbel et al. [7]	Domestic wastewater	Hybrid constructed wetland	Mixed macrophytes	3–10 days	High removal of COD, TSS, and nutrients	Hybrid systems improve treatment robustness
Nguyen et al. [13]	Surface/domestic wastewater	Vertical-flow constructed wetland	Vetiveria zizanioides	5–7 days	Efficient removal of organic matter and suspended solids	Performance depends on hydraulic loading and aeration
Compaoré et al. [8]	Greywater	Horizontal subsurface-flow constructed wetland	Emergent macrophytes	4–6 days	Reduction of fecal indicators and turbidity	Suitable for decentralized greywater reuse
Maiga et al. [9]	Greywater	Constructed wetland	Mixed macrophytes	5–8 days	Improved effluent quality for non-potable reuse	Supports reuse-oriented wetland systems
Farruggia et al. [11]	Municipal wastewater	Full-scale constructed wetland	Various macrophytes	Long-term operation	Seasonal variation affected treatment performance	Long-term monitoring is recommended

Note: COD = chemical oxygen demand; TSS = total suspended solids; TDS = total dissolved solids.

4.11 Limitations and Future Work

This study has several limitations that should be considered when interpreting the findings. First, the experiments were conducted without replication, with each treatment evaluated only once. Consequently, the results should be interpreted as preliminary trends rather than statistically validated conclusions, as treatment variability and reproducibility could not be assessed. Therefore, this study should be regarded as a pilot investigation intended to evaluate the feasibility and preliminary performance of a vetiver-based constructed wetland for greywater treatment. Future studies should incorporate appropriate experimental replication and statistical analyses to validate the observed treatment trends and improve the robustness of the conclusions.

Second, although the influent greywater originated from the same fish pond and was mixed before each experimental run, natural variations in greywater composition may still have occurred and were not continuously monitored. Such fluctuations in influent quality may have affected treatment efficiency. Future investigations should include systematic monitoring of influent characteristics, standardized homogenization procedures, and replicated sampling to better quantify variability and improve the reliability of treatment performance assessments.

The increase in nitrate and nitrite concentrations observed under certain experimental conditions suggests that nitrification may have occurred within the constructed wetland. As organic matter decreased and hydraulic retention time increased, oxygen-rich microenvironments likely developed around vetiver roots and within the filter media, promoting the oxidation of reduced nitrogen compounds. However, because dissolved oxygen, oxidation–reduction potential (ORP), and ammonium concentrations were not measured, this interpretation remains a plausible hypothesis rather than a confirmed mechanism. Future studies should directly monitor these parameters to better understand

nitrogen transformation pathways within the wetland system.

The results also demonstrated that treatment performance generally improved with increasing hydraulic retention time and higher vetiver planting density. Nevertheless, several effluent quality parameters, particularly turbidity, colour, and fecal coliform, did not consistently comply with the applicable treated effluent standards. Consequently, the treated water should not be considered suitable for potable use or other applications involving direct human contact. Instead, it may be appropriate for restricted non-potable reuse, such as landscape irrigation, irrigation of non-food crops, or other applications permitted under local regulations. Where higher water quality is required, additional treatment processes, including filtration, sedimentation, or disinfection, should be incorporated. These findings indicate that the proposed vetiver-based constructed wetland is more suitable as a pre-treatment or polishing unit than as a standalone treatment system for producing high-quality reclaimed water.

Finally, the present investigation was conducted under laboratory-scale conditions over a relatively short experimental period. Consequently, the influence of seasonal variation, long-term hydraulic performance, media clogging, plant growth dynamics, and operational stability could not be evaluated. Future research should therefore include pilot-scale and full-scale field investigations with extended monitoring periods, repeated experiments, comprehensive water quality analyses, and long-term performance evaluations. Such studies will provide a more reliable basis for optimizing system design and supporting the wider implementation of vetiver-based constructed wetlands for sustainable greywater treatment and reuse.

5 Conclusions

Greywater generated from tilapia pond operations contains organic matter, suspended solids, nutrients, and microbial contaminants that require appropriate treatment before being discharged into receiving water bodies. This study evaluated the performance of a laboratory-scale constructed wetland planted with *Vetiveria zizanioides* (vetiver grass) using submerged soil media under different hydraulic loading rates, planting densities, and hydraulic retention times.

The results demonstrated that the vetiver-based constructed wetland effectively improved greywater quality. The highest treatment performance was achieved with 150 vetiver stems and a hydraulic retention time of seven days. Under these conditions, the treated effluent met the applicable treated effluent quality standards for pH (7.90), nitrate (2.319 mg/L), nitrite (0.0922 mg/L), iron (0.0286 mg/L), and TDS (175 mg/L). However, several parameters, including odor (0.77), colour (10.6 TCU), turbidity (5.44 NTU), and fecal coliform (85 CFU/100 mL), did not fully comply with the required treated effluent quality standards. These findings indicate that the proposed treatment system is effective in improving greywater quality but is more suitable as a pre-treatment or polishing unit rather than as a standalone treatment system for producing water suitable for unrestricted reuse.

The study also demonstrates that vetiver-based constructed wetlands represent a promising nature-based and low-energy technology for decentralized greywater treatment in aquaculture and small-community settings. In addition to reducing pollutant concentrations, the system has the potential to decrease contaminant discharge into surrounding aquatic environments, thereby supporting more sustainable water resource management.

Nevertheless, this study has several limitations. The experiments were conducted under laboratory-scale conditions without experimental replication, and the influent water quality was not continuously monitored throughout the study. Consequently, the findings should be regarded as preliminary and should not be directly extrapolated to full-scale applications. Future research should include replicated experiments, continuous influent monitoring, pilot- and field-scale investigations, long-term performance evaluation under seasonal conditions, and additional treatment processes, such as filtration or disinfection, to improve microbial removal and expand opportunities for safe non-potable water reuse.

Author Contributions

Conceptualization, L.K.; methodology, L.K., W.M., and H.; investigation, L.K., W.M., H., M., and M.W.; formal analysis, L.K., H., and N.S.; data curation, L.K. and N.S.; visualization, N.S. and M.W.; writing—original draft preparation, L.K.; writing—review and editing, W.M., H., M., M.W., and N.S.; supervision, W.M.; project administration, L.K.; funding acquisition, L.K. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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